FNCE 30007 EXAM PRACTICE QUESTIONS

QUESTION 1

This question requires you to consider a three-asset portfolio valued at 10 million AUD. The portfolio consists of the following assets: AMP, Commonwealth Bank (CBA) and QBE. The variance covariance matrix of 5 day continuously compounded returns is equal to

	AMP	CBA	QBE
AMP	0.000917	0.000426	0.000469
CBA	0.000426	0.000558	0.000316
QBE	0.000469	0.000316	0.001557

(a) Define the Value at Risk (VaR) for a portfolio.

(b) Assuming portfolio weights of AMP (40%), CBA(30%), QBE(30%), calculate the 99% 5 day relative VaR estimate (employ a z score measured to 2 decimal places)

(c) Calculate the VaR diversification benefit of the portfolio.

QUESTION 2

Consider a European call on a non-dividend paying stock with a strike of $50, maturity two weeks, current stock price of $52, risk free rate of 5% p.a and volatility of 20% p.a. Assume a bank has just sold an option for $3 and wants to hedge the exposure dynamically at the end of each week. The BSM valuation for the option is $2.2569.

Complete the following table (to 4 decimal places).

				Bond position				
End of week	S_t	Δ	# shares bought/sold	Open	Interest	Shares	Option	Close
0	52	0.8577		0	0			
1	48							
2	35							

QUESTION 3

a) Explain the concept of the riskless hedge and how it can be used to price an option.

b) What is risk neutral valuation? How does it relate to the riskless hedge approach to option pricing?

QUESTION 4

Consider an American put option with a strike of $9.50, a spot price of $10, volatility of 20% p.a, risk free rate of 5% p.a., time to maturity of 3 months and a dividend that is paid at 5% of the stock value in 2 months. T Use a three period binomial model to price the option (assuming the ex-div drop off occurs at t=2 months).

QUESTION 5

Consider a put option on the ASX200 index. Assume the index currently stands at 5,768. It is expected to increase or decrease by 15% over each of the next two time periods of two months. The risk-free rate is 5.75% and the dividend yield on the index is 2.5%.

a) What is the value of the option if it is European with four-months to maturity and has an exercise price of 5,700. Show your calculations.

b) What is the value of the option if it is American with four-months to maturity and has an exercise price of 5,700. Show your calculations.

QUESTION 6

Assume that the spot price of Swiss Franc is U.S. $1.05 with a volatility of 7% per annum. The risk-free rates in Switzerland and the U.S. are 3% and 7% per annum. Assume that the U.S. is the home market.

a) Determine the value of a European call option to buy one Swiss Franc for U.S. $1.05 in seven months. Show your calculations.

b) What is the price of a European put option to sell one Swiss Franc for U.S. $1.05 in seven months? Show your calculations.

c) Find the price of a call option to buy U.S. $1.05 with one Swiss Franc in seven months? Show your calculations.

QUESTION 7

Consider a European call option on a non dividend paying stock when the stock price is $25.00, the strike price is $28.00, the risk-free interest rate is 8% per annum, the volatility is 30% per annum and there is four years to maturity.

a) Find the current price of the option. Show your calculations.

Now assume the stock price instantaneously changes to $25.50.

b) Use the delta of the option to estimate the value of the option after the change. Show your calculations.

c) Use the delta and gamma of the option to estimate the value of the option after the change. Show your calculations.

d) What is the exact value of the option after the change? Show your calculations.

QUESTION 8

Find the price of a European put option on a dividend paying stock when the stock price is $80, exercise price is $90, continuously compounded risk-free interest rate is 9% per annum, volatility is 2.52% per trading day and there is 7 months to maturity. Assume that the stock is expected pay a dividend of $1.2 in the next 3, 6, and 9 months. Show your calculations.

QUESTION 9

Assume the spot USD/JPY exchange rate is 96 (1 USD = 96 JPY). The continuously compounded risk-free interest rate is 1% per annum in the US and 3% per annum in Japan. Assume that the volatility of the USD/JPY exchange rate is 25% per annum.

a) Find the current price of JPY value of a one-year European call option on one USD with an exercise price of JPY 90? Show your calculations.

b) Calculate the USD value of a one-year European put option on one JPY with an exercise price of USD 0.011. Show your calculations. (*Hint: there is no need to use the Black-Scholes*)

QUESTION 10

Assume you are seeking to price a stock option using monte-carlo techniques. The following discretised geometric Brownian motion is the assumed underlying data generating process

$$S_t = S_{t-1} \exp\left[\left(rf - dy - 0.5\sigma^2\right)dt + \sigma\sqrt{dt}v_t\right]$$

where S_t is the stock price at time t, rf is the risk free rate and dy is the dividend yield. Assume the current stock price S_t =\$100, rf = 5% p.a, dy = 2% p.a. and σ = *0.2.* Furthermore assume that dt (the change in time) represents one day dt = 0.004.

a) Assume that you are seeking to simulate prices over the next two days. The first replication draws the following random numbers (from the normal distribution): v_{t+1} = 2.5 and v_{t+2} = -1.1. Calculate the simulated stock prices for periods $t+1$ and $t+2$.

b) What are antithetic variates? Why are they used?

c) Assume the second replication simulates prices using antithetic variates. What is the simulated spot price at time $t+1$?

d) How would you use the simulated prices to value a European call?

QUESTION 11

Consider an **American call** option on one share. The current stock price is \$100 and the stock is expected to pay a dividend of \$8.84 per share in 1 years time. The strike price of the option is \$110, the risk free interest rate is 10% per annum, the volatility is 20% per annum and there is 2 years to maturity. What is the current value of the option using Black's (1975) Pseudo American option pricing model ?